

COMP 9200

Applied Research Methods

Getting started

Land acknowledgment

The British Columbia Institute of Technology acknowledges that our campuses are located on the unceded traditional territories of the

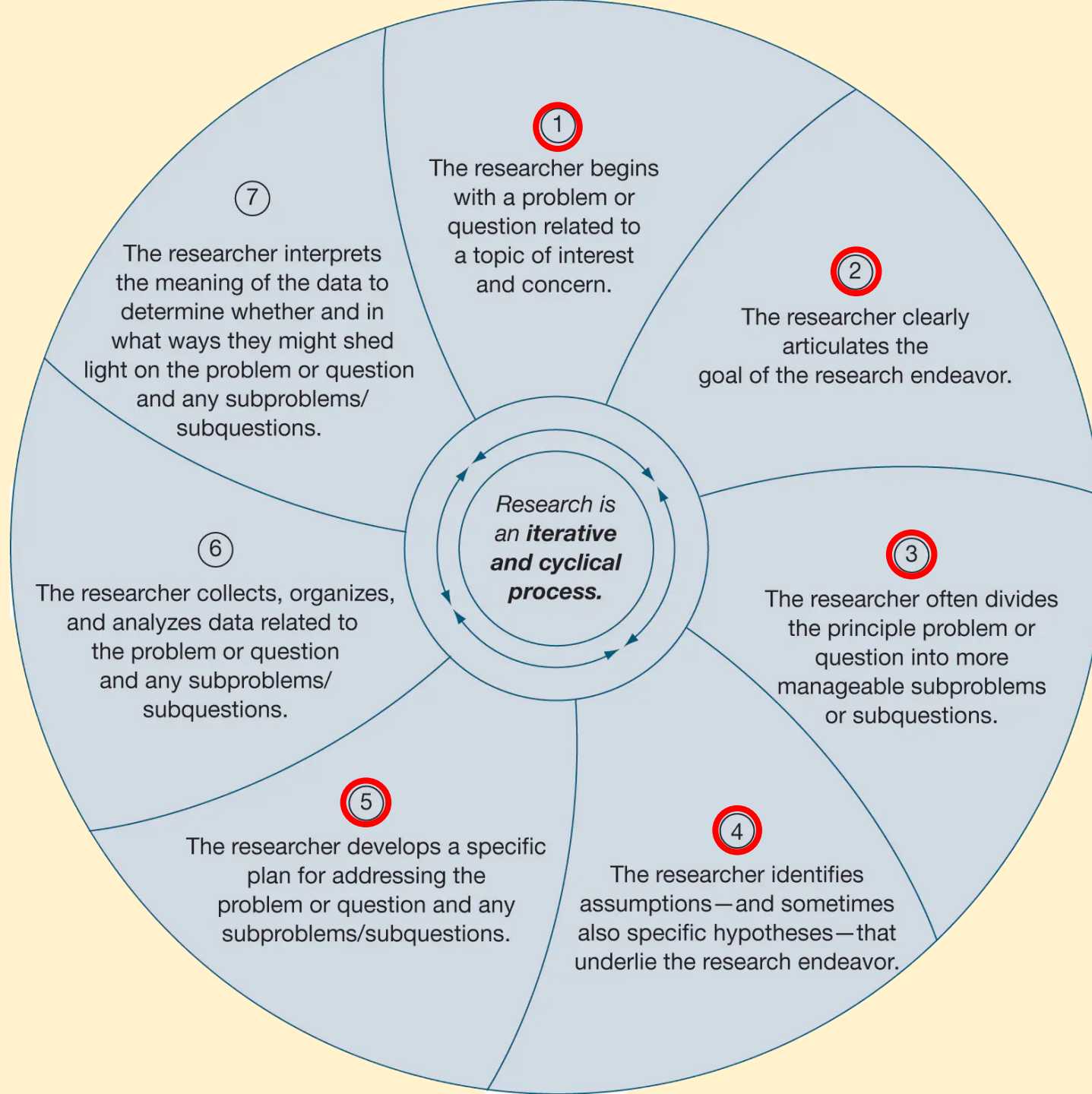
Coast Salish Nations of

Skwxwú7mesh (Squamish),
səl'ilwətaʔt (Tsleil-Waututh), and
xwməθkwəyəm (Musqueam)

Overview

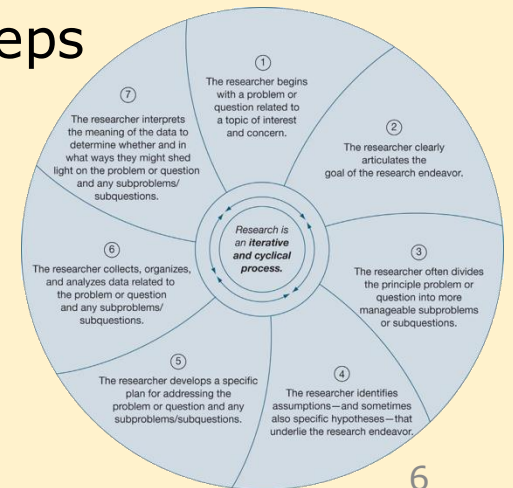
- Nature of research
- Defining the research
- The research proposal
- Academic writing basics
- Reviewing other works

Nature of research



What Research Is

- Systematic process of collecting, analyzing, and interpreting information—data
 - Formal research: to increase our understanding of a phenomenon and communicate what we discover
- Research cycle (generally involving seven distinct steps)
 - Iterative process
 - Researcher moves back and forth between 2 or more steps
 - Cyclical process
 - Step 7 is usually not the final step



What Research Is Not

- Not merely gathering information
 - Not to be viewed as information discovery or reference skills
- Not merely rummaging around for hard-to-locate information
 - Not to be viewed as an exercise in self-enlightenment
- Not merely transporting facts from one location to another
 - Not to be viewed as fact documentation and organization

The human mind

- Critical thinking: verbal reasoning, argument analysis, probabilistic reasoning, decision making, hypothesis testing
- Deductive logic: reasoning that begins with premises or assumptions and proceeds toward conclusions
- Inductive reasoning: process beginning with observations and proceeding to broader generalizations
- Scientific method: systematic and rigorous yet flexible process
- Theory building with a scientific attitude: well-documented findings; logically defensible reasoning (may use abduction)
- Collaboration with other minds

The human mind – some pitfalls

1. Confusing what must logically be true with what seems to be true (deductive reasoning pitfall)
2. Making generalizations about members of a category based on a restricted subset of the category (inductive reasoning pitfall)
3. Looking only for evidence that supports our hypotheses
4. Confirming expectations even in the face of contradictory evidence
5. Mistaking dogma for fact
6. Letting emotions override logic and objectivity
7. Mistaking correlation for causation

Defining the research

Finding & defining the problem

Basic research

- Enhance basic knowledge about physical, biological, psychological or social world
- Shed light on historical, cultural or aesthetic phenomena

Applied research

- Address issues with immediate relevance to current practices, procedures or policies
- Make decisions about practical problems
- Address questions in one's immediate work environment (action research)

Finding & defining the problem

Criteria:

- Address important question
 - answer will make a difference
- Advance frontiers of knowledge
 - lead to new ways of thinking
 - suggest possible applications
 - pave way for further research

Finding & defining the problem

Things to avoid:

- projects that are ruse for achieving self-enlightenment
- problem whose sole purpose is to compare two sets of data
- calculating correlation coefficient between two sets of data to show relationship between them
- problems that result in yes/no answer

Finding & defining the problem

Ways to find legitimate problem:

- Look around you
- Read the literature
- Attend professional conferences
- Seek advice of experts
- Choose topic that intrigues and motivates you
- Choose topic that others will find interesting and worthy of attention

Finding & defining the problem

Stating the problem:

- State problem clearly and completely
- Think through feasibility of project that problem implies
- Say precisely what you mean
- Absolute honesty and integrity are the rule!
- State problem so it reflects an open mind about its solution
- Edit your work

Subproblems

Subproblem:

- Subparts of main problem
- Integral part of main problem
- Pseudo-subproblems:
 - Procedural issues
 - Involve decisions that must be made before resolving problem and subproblems
- Completely researchable unit
- Clearly tied to interpretation of data
- Add up to totality of problem
- Small in number

Define Terms

- Must precisely define terms in problem and subproblems
- Define each term as it will be used in project
- Make term mean whatever you wish it to mean within context of problem and its subproblems

State Assumptions

- Assumptions are basic to research problem
- All assumptions with material bearing on problem should be openly and unreservedly set forth
- Statement of assumptions necessary for others to evaluate conclusions
- Reveals what researcher may be taking for granted with respect to problem

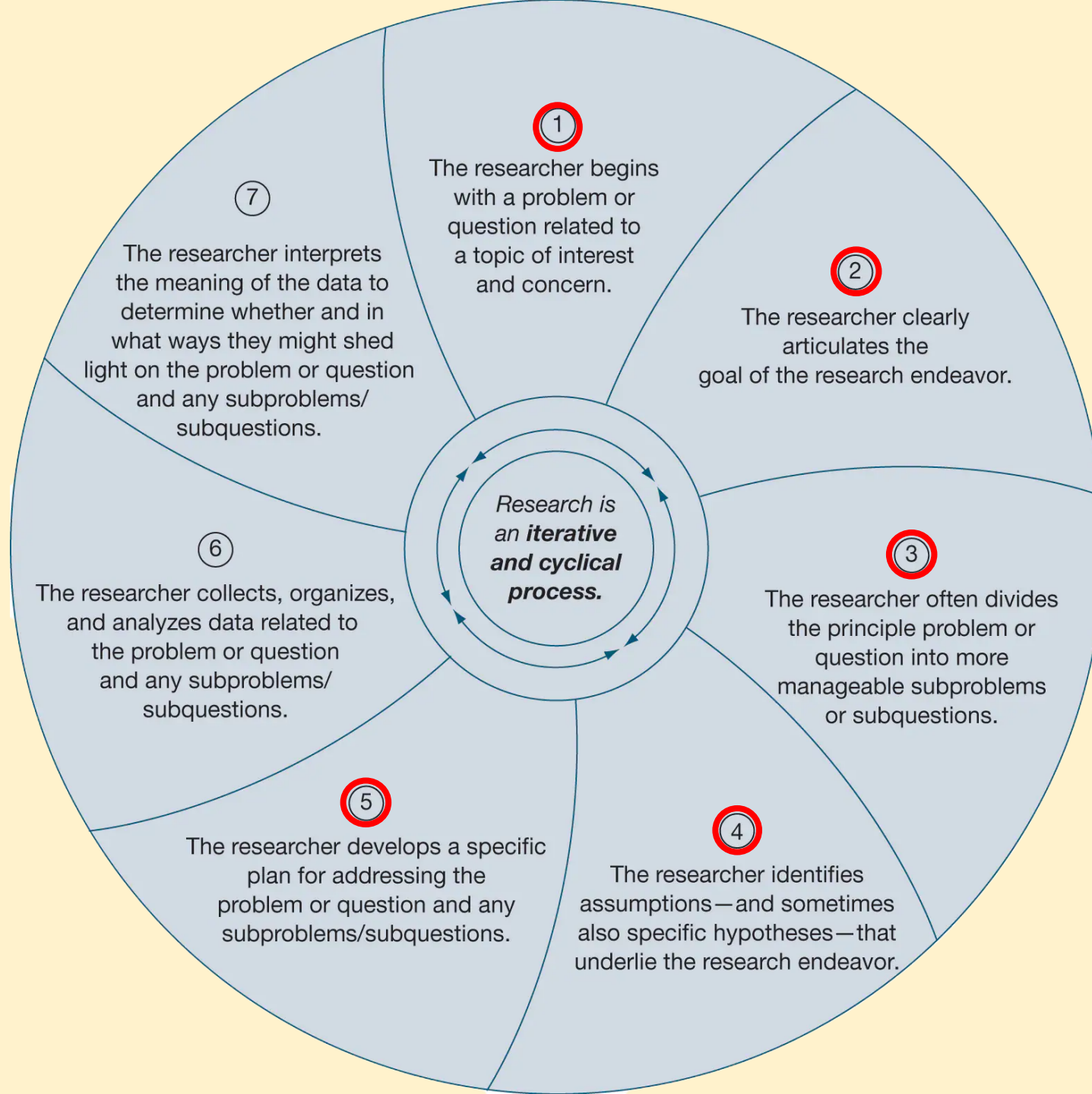
Hypothesis & research question

- Hypotheses essential to experimental research
- Research questions more common in qualitative research
- Both provide guidance for data to be collected
- Both suggest how data should be analyzed and interpreted
- Both may originate in subproblems
- Provide position from which researcher may initiate exploration of problem
- Act as checkpoints against which to test findings that data reveal

Hypothesis & research question

Delimit research:

- What you are not going to do
- Avoid data extraneous to research problem
- Stick with what is relevant to research problem



Finding articles

- BCIT Library
 - Indexes
 - Journals
 - Books
 - Conference papers
- Google Scholar
- Wikipedia
- Patents

Research proposal

Writing a Proposal: Overview

- State the subproblems
- Write the hypotheses/questions
- Write the delimitations
- Write the definitions of terms
- Write the assumptions
- Describe the importance of the study
- Type the proposal

Writing a Proposal

- Have you conducted thorough literature search to justify time and effort expended on your research project?
- Have you looked at your research problem from all angles to minimize unwanted surprises?
- What research procedures will you follow?
- What research tools are available?
- Can others read/understand the proposal?

Fine-tuning Research Problem

- Conduct a thorough literature review
- Try to see the problem from all sides
- Think through the process
- Use all available tools and resources
- Discuss your research problem with others
- Hold up your proposed project for others to examine and critique
- Remember that your project will take a great deal of time

General criteria

- Universality
 - project should be one that might be carried out by any competent person
 - researcher = catalyst who collects, organizes, and reports what collected data seem to indicate
- Replication
 - research should be repeatable
 - any other competent person should be able to take the problem and, collecting data under same circumstances and within same parameters, achieve comparable results

Academic writing basics

Academic writing

Be specific and precise

- Choose words and phrases carefully for exact meaning
- Say what you mean and mean what you say

Clear objective and focus

- What is the main objective?
- Every source and statement: how does it relate?

Academic writing

Organize thoughts and ideas

- Flow of ideas
- Use categories and subcategories (headings and subheadings)
- In some ways, you are telling a story (or outlining your story in the case of the proposal)
- But the story should have a direction, not aimless or musings

Academic writing

Use overview

- What should reader expect?
- Example from textbook:

The first part of this review will examine the theories, frameworks, and experimental research behind the research on adjunct questioning. Part two will investigate the transition of adjunct questioning to self-generated questioning. Specific models of self-generated questioning will be explored, starting with the historical research on question position [and progressing] to the more contemporary research on individual differences in self-questioning. Part three will explore some basic research on note taking and tie note taking theory with the research on self-generated questioning. (Jackson, 1996, p. 17)

Academic writing

Use concrete examples especially for abstract ideas

- Abstract vs. vague
- Example from textbook:
- Abstract idea: *"how a researcher evaluates, or codes, the content of students' class notes will affect what the researcher discovers about those notes. More specifically, she argued that only a superficial coding scheme (e.g., counting the number of main ideas included in notes) would fail to capture the true quality of the notes."*
- Example: *"For example, while listening to the same lecture, Student A may record only an outline of the lecture, whereas Student B may record an outline, examples, definitions, and mnemonics. If a researcher only considered the number of main ideas that students included in their notes, then both sets of notes might be considered equivalent, despite the fact that the two sets differ considerably in the type of material recorded. (Jackson, 1996, p. 9)"*

Academic writing

Use figures and tables

- Figures: well-annotated, captioned, referenced
- Tables: easy to read, captioned, referenced

Academic writing

Summarize each section

- Should roughly mirror the overview
- Highlight most important points and flow of ideas
- Example from textbook:

In sum, the students understand attention to be a mental process. They know their attention is inconsistent and affected by emotions and interest. They also realize that the right level of material, amount of information, and length of time helps their attention. The stillness of reading is difficult for some of the students but calming for others, and they appear to know this, and to know when reading will be difficult and when it will be calming. This idea is contrary to what has been written in the literature about struggling readers. (Zambo, 2003, p. 68)

Academic writing

Anticipate writing multiple drafts

- For yourself:
 - writing ideas different than thinking or talking
 - reading what you wrote helps clarify your own thinking iteratively
- For others:
 - what makes perfect sense to you does not to others
 - once someone has seen a draft, can be biased, so having a few to look at it helps
- Helps critically evaluate:
 - logic and ideas
 - precision and clarity of expression
 - gaps or errors

Grammar and spelling

- First line of defense:
 - Use built-in spelling and grammar checkers on MS Word (actually most modern word processors)
- Second line of defense:
 - Use BCIT writing resources (more later)
- Third line of defense:
 - Have a friend or family member proofread your work!

Grammar and spelling

Incorrect:

Watch you're words! Spell-check may not see words that are miss used because they are spelled rite!

Correct:

Watch your words! Spell check may not see words that are misused because they are spelled right!

Clarity and style

- Check for things such as:
 - Run-on sentences

The project will use TensorFlow, it will be used as the main library.

The project will use TensorFlow. It will be used as the main library.

OR

The project will use TensorFlow as the main library.

Clarity and style

- Check for things such as:
 - Sentence fragments

This project, which will enhance my expertise in my chosen specialization, which is HCI.

This project will enhance my expertise in my chosen specialization of HCI.

Clarity and style

- Check for things such as:
 - Missing articles, etc.

The game use ANN as algorithm for enemy behaviour.

The game will use ANN as the algorithm for controlling the enemy's behaviour.

Clarity and style

- Check for things such as:
 - Writing the way you speak, contractions, etc.

I don't intend to do too much optimization.

Optimizing the code will generally be outside the scope of the project.

Grammar and spelling

- Other helpful links:
- <https://authority.pub/common-grammar-mistakes/>
- <https://www.inc.com/melanie-curtin/11-common-writing-mistakes-to-avoid.html>
- <https://owl.excelsior.edu/grammar-essentials/common-errors/>

BCIT Writing Resources

- <https://commons.bcit.ca/library/2018/10/17/6453/>
- <https://www.writeaway.ca>

Reviewing other works

Reviewing articles

- What is the merit?
- Is it presented well?
- Does it have a logical structure?
 - Abstract
 - Introduction
 - Body (technique, results, discussion)
 - Conclusions
 - References
 - Tables and figures (with captions)

Reviewing articles

- Abstract
 - Summarizes the paper
 - Includes conclusions as well as statement of original problem
- Introduction
 - Explains why the topic is important
 - Reviews other work

Reviewing articles

- Body
 - Approach and analysis
 - Results: basis for claims
 - Errors and uncertainties
 - “Right” level of detail
- Conclusions
 - Summarizes – no surprises or new information
 - Points to future work (separate section?)

Reviewing articles

- References
 - Other work
 - Especially important for more recent work
- Tables and figures
 - Conveys information and findings visually
 - Describes design and concepts

Tips

- Search for generic terms, and get a sense for which publications and authors seem to be publishing in the area
- Look for articles that are cited a lot, or look for common references in articles
- Then look at articles that reference popular articles

Review

- Nature of research
- Defining the research
- The research proposal
- Academic writing basics
- Reviewing other works